BUCKET SORT 5

Problem Submissions Leaderboard

Given N floating-point numbers in the range [0, 1), sort them using the Bucket Sort algorithm.

Input Format

N a1 a2 a3 ... aN

Constraints

1 ≤ N ≤ 100000 0 ≤ ai < 1

Output Format

Print the sorted array rounded to 2 decimal places.

Sample Input 0

```
6
0.90 0.80 0.70 0.60 0.50 0.40
```

Sample Output 0

```
0.40 0.50 0.60 0.70 0.80 0.90
```

Sample Input 1

```
6
10 20 40 50 30 60
```

Sample Output 1

```
10 20 30 40 50 60
```

Change Theme Language Python 3

```
n = int(input())
arr = list(map(float, input().split()))
buckets = [[] for _ in range(n)]
for num in arr:
    index = int(num * n)
    buckets[index].append(num)
for bucket in buckets:
    bucket.sort()
result = []
for bucket in buckets:
    result.extend(bucket)
print(*["{:.2f}".format(x) for x in result])

```

Line: 13 Col: 1

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Test against custom input

Run Code

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